



COURSE SLIDES

# AWS Certified Machine Learning Engineer Associate Training

TGS-2024049340 | Hands-On Labs | Tertiary Infotech Academy Pte Ltd

Version 2

# Copyright Notice

 Copyright 2026 Tertiary Infotech Academy Pte Ltd.

 Materials are for enrolled learners only.

 Do not redistribute assessment or trainer materials.

# About Tertiary Infotech Academy

 WSQ course provider delivering ICT and business technology training.

 UEN: 201200696W

 Training format: instructor-led, hands-on, assessment-ready.

# About the Trainer



## Your Trainer

General Trainer template -  
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

# About the Trainer



**Dr. Alfred Ang**

Principal Trainer  
Tertiary Infotech Academy Pte. Ltd.

## ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

## QUALIFICATIONS

PhD - specialises in AI, automation, software engineering and cloud learning.

## DELIVERS

WSQ courses on cloud fundamentals, AI, automation and app development.

## FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

# Learner Introduction

-  Introduce yourself and your current role.
-  Share your AWS and machine-learning experience.
-  Name one production ML challenge you want to solve.

# Digital Attendance

 Scan SSG digital attendance at required checkpoints.

 Minimum 75% attendance is required.

 Use the same registered learner identity throughout.

# Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.



# Course Objectives

Data

## Labs 1-2

Secure, ingest, transform and  
validate ML data

Model

## Labs 3-5

Select, train, tune, evaluate  
and register

Deploy

## Labs 6-7

Inference infrastructure and  
MLOps automation

Operate

## Lab 8

Monitoring, security, cost and  
exam readiness

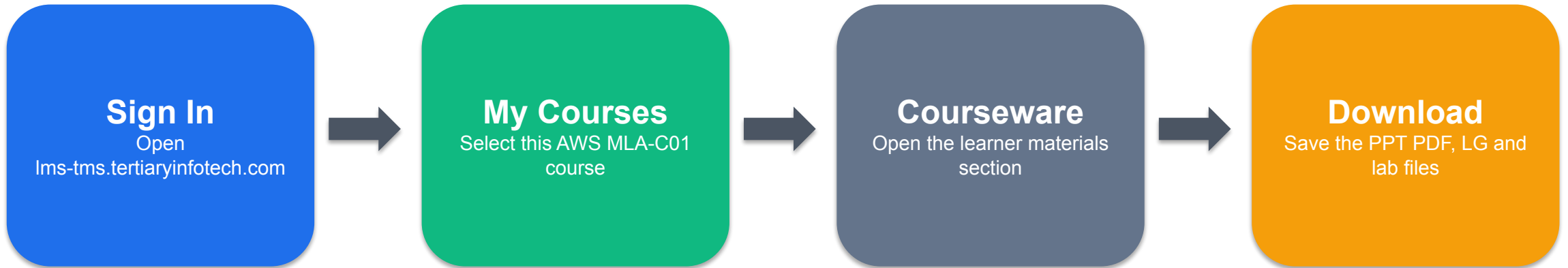
# Learning Resources

 AWS Management Console: <https://console.aws.amazon.com>

 AWS documentation: <https://docs.aws.amazon.com>

 Official AWS Certified Machine Learning Engineer - Associate exam guide.

# Download Course Material



Courseware and assessment submission are handled through <https://lms-tms.tertiaryinfotech.com/>.

# Briefing for Assessment

- The assessment is individual and open book.
- WA tests knowledge taught in the deck and Learner Guide.
- PP follows the Lab 1-8 workflow and requires evidence.
- Complete both digital attendance checkpoints and submit through the LMS.

# Assessment

 Written Assessment (WA): 1 hour.

 Practical Performance (PP): 1 hour.

 Assessment starts at 4:00 PM on Day 2 and ends at 6:00 PM.

# Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

# Take the Online AWS MLA-C01 Practice Exams

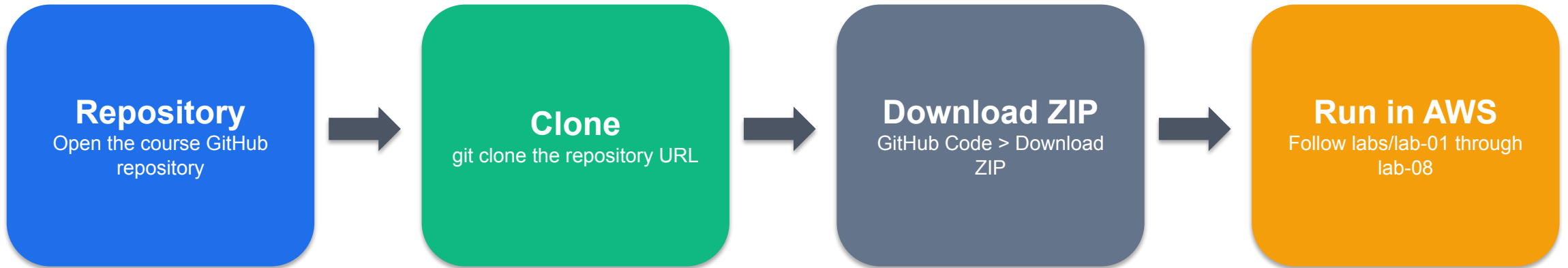
- Reinforce AWS MLA-C01 domains with realistic, exam-style questions before assessment day.
- Use practice sets to test data preparation, model development, deployment, MLOps and monitoring.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to review.

**Access and practise at:**

<https://exams.tertiaryinfotech.com/>

The screenshot shows the Tertiary Exams website. At the top, there is a navigation bar with a blue circular logo, the text 'Tertiary Exams', and links for 'Practice Exam' and 'About Us'. A blue 'Get started' button is in the top right corner. Below the navigation bar, the main heading is 'AWS Machine Learning Engineer - Associate Practice Exams'. Under this heading, there are three white boxes with colored borders: 'MODE' with a blue border and the text 'Practice', 'FOCUS' with a green border and the text 'AWS MLA-C01', and 'ACCESS' with a purple border and the text 'Class link'. To the right of these boxes is a 'PRACTICE DETAILS' box containing the text: 'Exam code AWS MLA-C01', 'Use Review', and 'Domains 3'.

# Access the Hands-On Labs



<https://github.com/tertiarycourses/TGS-2024049340---AWS-Certified-Machine-Learning-Engineer-Associate-Training>



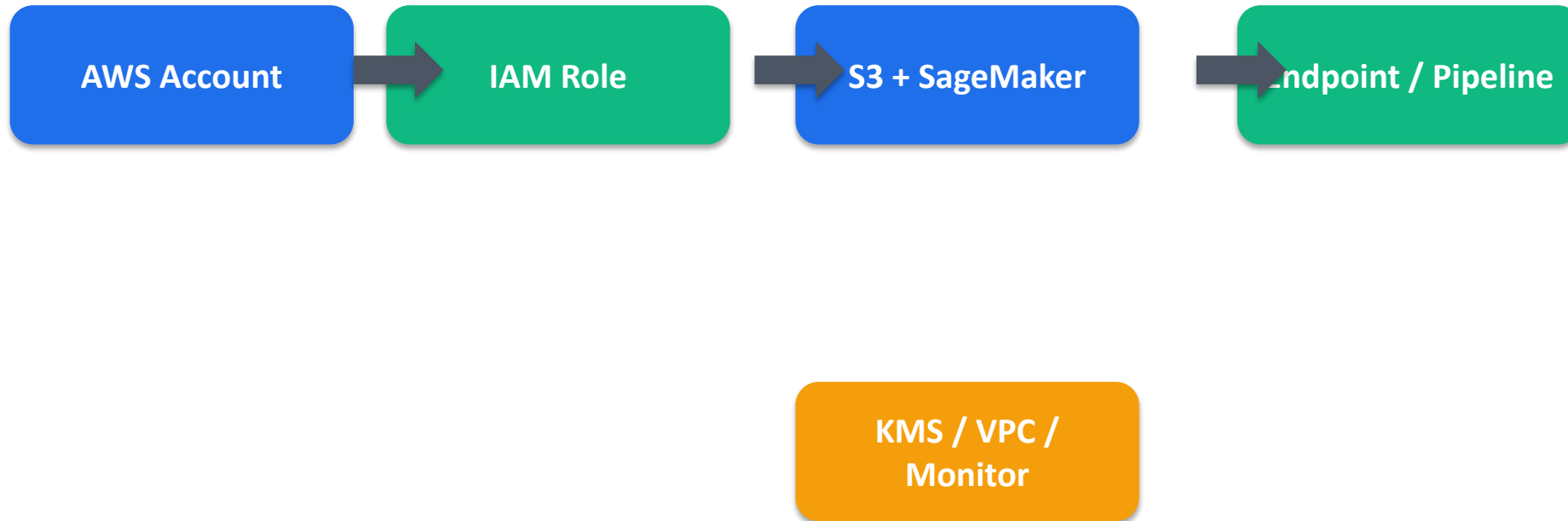
# Two-Day Lesson Flow

● Day 1: Labs 1-4 - account safety, data preparation, modeling, training and tuning.

● Day 2: Labs 5-8 - evaluation, deployment, MLOps, monitoring, security and cost.

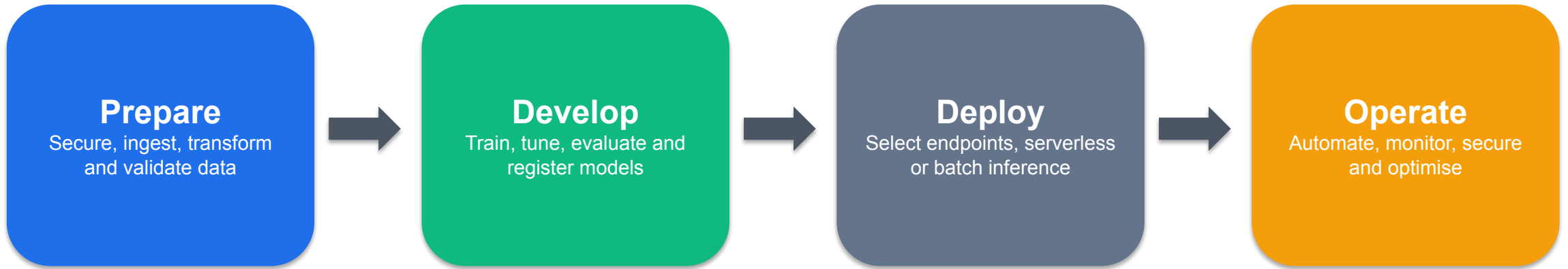
● Complete each validation and cleanup step before moving to the next lab.

# AWS ML Workload Resource Model



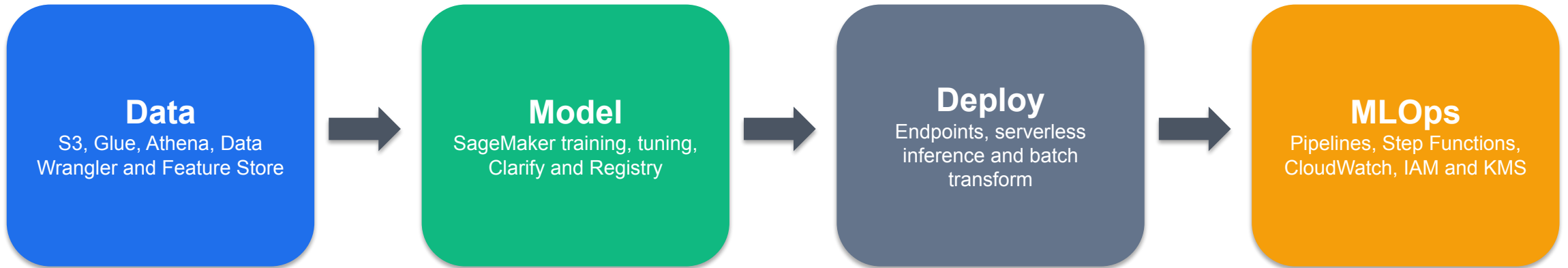
The labs progress from account and IAM safeguards to governed data, SageMaker development, deployment automation, monitoring, security and cost control.

# AWS ML Engineering Lab Flow



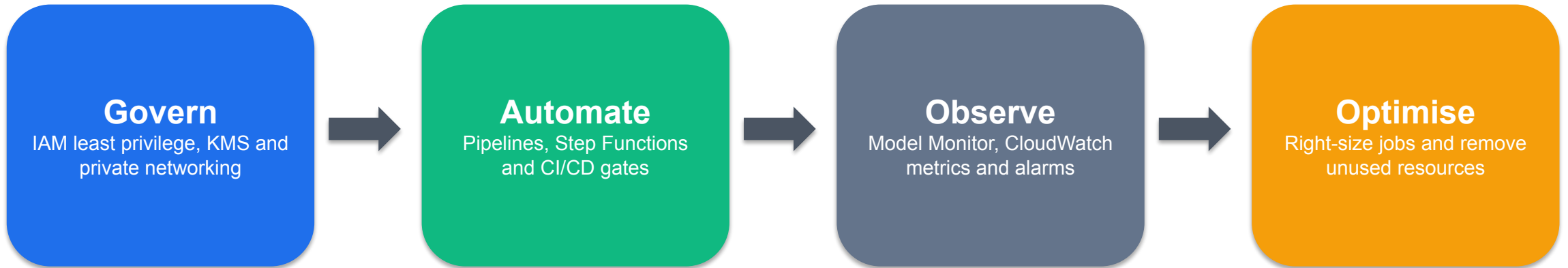
The slide deck, Learner Guide and Lesson Plan follow this same lab sequence.

# AWS Services Practised



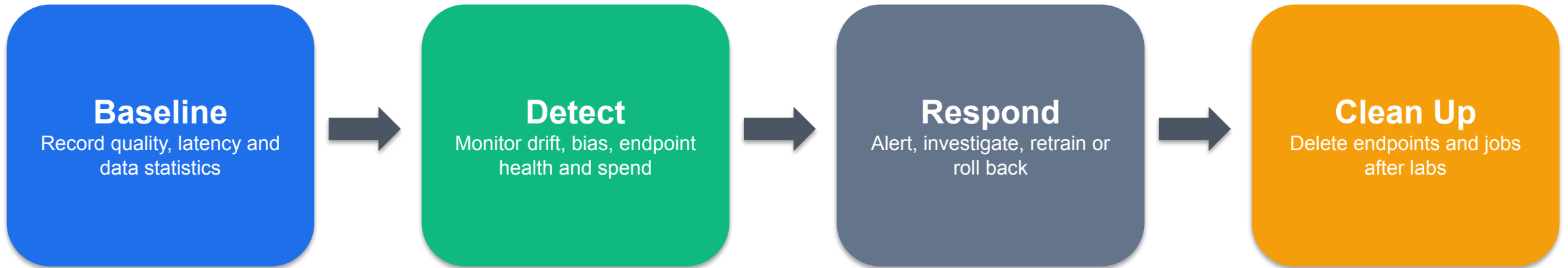
The service groups map directly to the eight labs and the MLA-C01 task domains.

# Management and Governance Loop



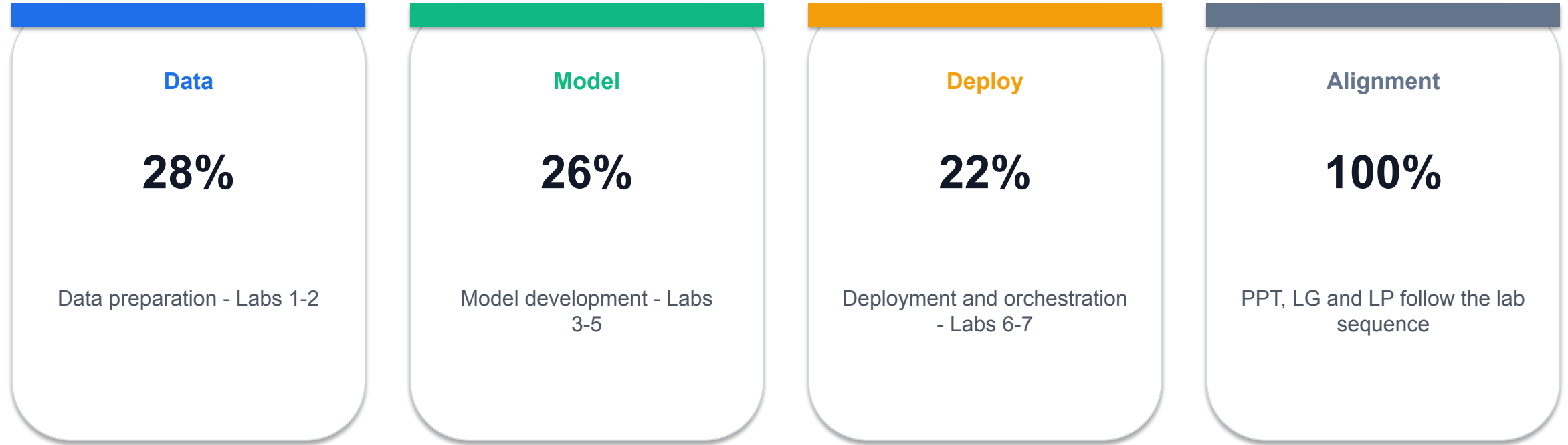
The Day 2 labs turn resources into governed cloud assets through identity, access, policy and monitoring.

# Cost and Monitoring Lifecycle



Cost estimation, budgets, Advisor, Service Health, Monitor and cleanup are handled as one operating loop.

# AWS MLA-C01 Domain Map



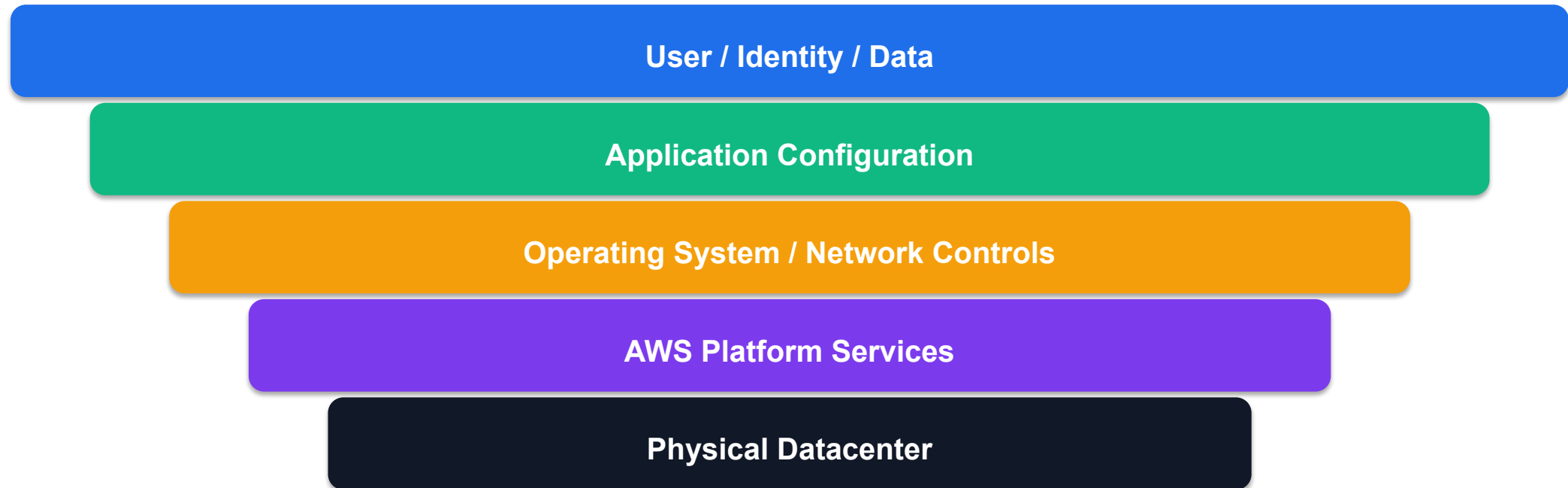
# AWS MLA-C01 Domain Weight Matrix

| Domain            | Exam Weight | Labs | Practice Priority |
|-------------------|-------------|------|-------------------|
| Data Preparation  | Labs 1-2    | 1-2  | High              |
| Model Development | Labs 3-5    | 3-5  | High              |
| Deploy & Operate  | Labs 6-8    | 6-8  | High              |

This matrix mirrors the CySA+ risk-matrix layout, adapted to AWS MLA-C01 domain coverage and lab priorities.



# AWS Shared Responsibility Layers



Learners configure the top layers in the labs; Microsoft operates the lower platform and physical layers.

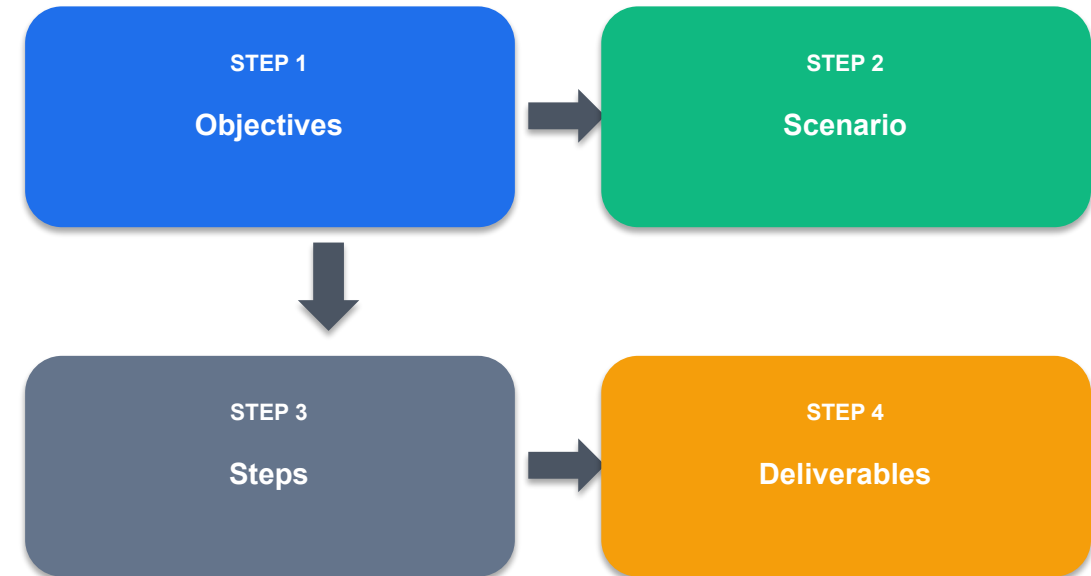
# Lab 1 - Account Safety, IAM, S3 ML Data Lake, and Data Formats

## Domain 1 - Data Preparation for Machine Learning

Day 1

labs/lab-01-account-i  
am-s3-ml-data-lake.  
md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

# Lab 1 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

# Lab 1 - Verify and Clean Up

## Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

## Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

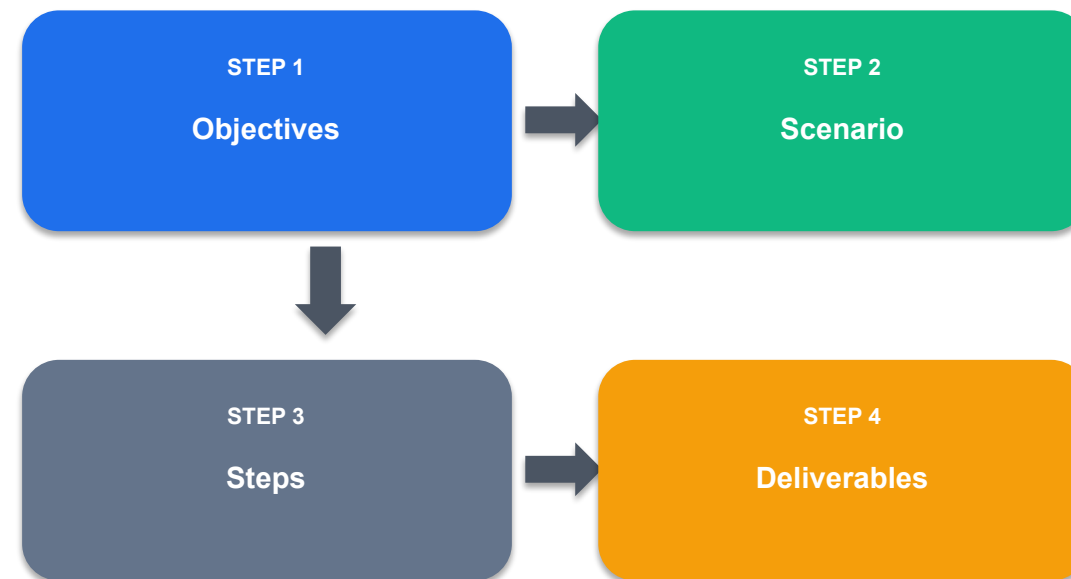
# Lab 2 - Data Ingestion, Transformation, Feature Engineering, and Data Quality

## Domain 1 - Data Preparation for Machine Learning

Day 1

labs/lab-02-data-ingestion-feature-engineering-quality.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

## Lab 2 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

## Lab 2 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

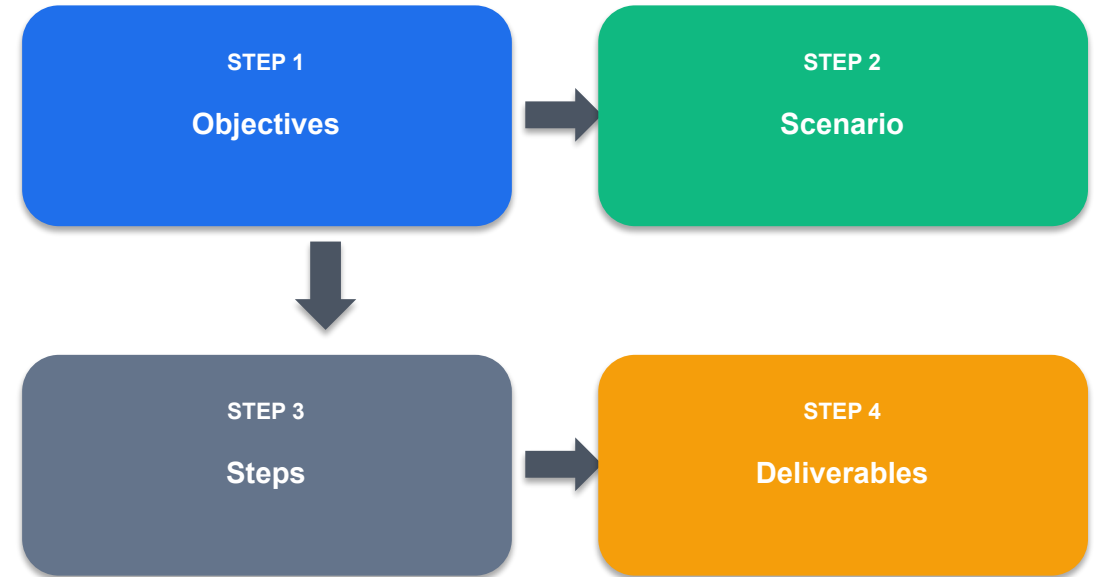
# Lab 3 - Modeling Approach, SageMaker Algorithms, JumpStart, and AI Services

Domain 2 - ML Model Development

Day 1

[labs/lab-03-modeling-approach-sagemaker-ai-services.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.



## Lab 3 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

## Lab 3 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

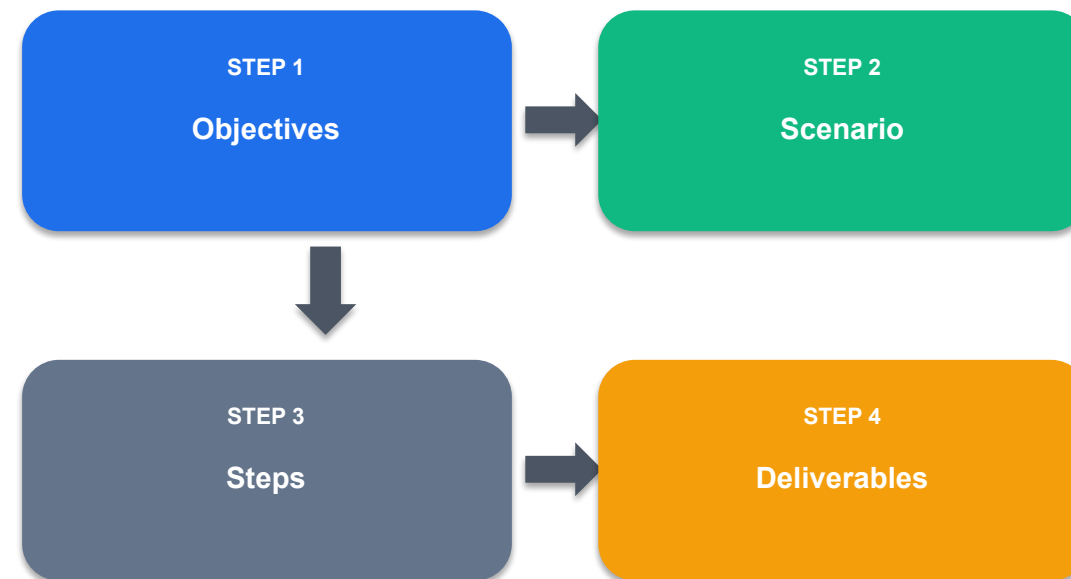
# Lab 4 - Training, Hyperparameter Tuning, and Model Registry

## Domain 2 - ML Model Development

Day 1

[labs/lab-04-training-tuning-model-registry.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

## Lab 4 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

## Lab 4 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

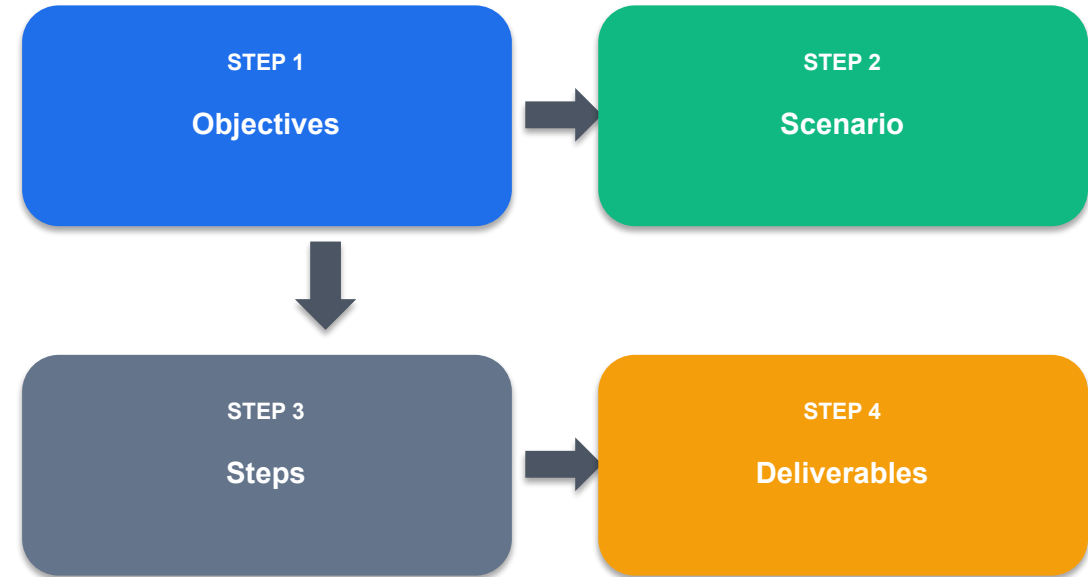
# Lab 5 - Model Evaluation, Performance Metrics, and Bias Analysis

Domain 2 - ML Model Development

Day 2

labs/lab-05-model-evaluation-performance-bias.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

## Lab 5 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

## Lab 5 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.



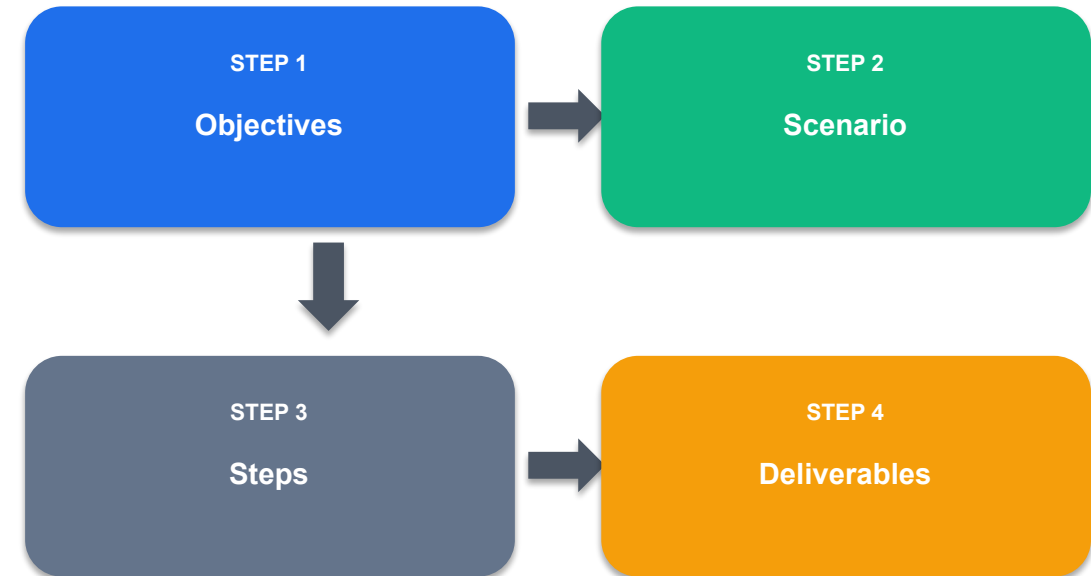
# Lab 6 - Deployment Infrastructure, Endpoints, Batch Transform, and Auto Scaling

Domain 3 - Deployment and Orchestration of ML Workflows

Day 2

labs/lab-06-deployment-endpoints-inference-infrastructure.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

## Lab 6 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

## Lab 6 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

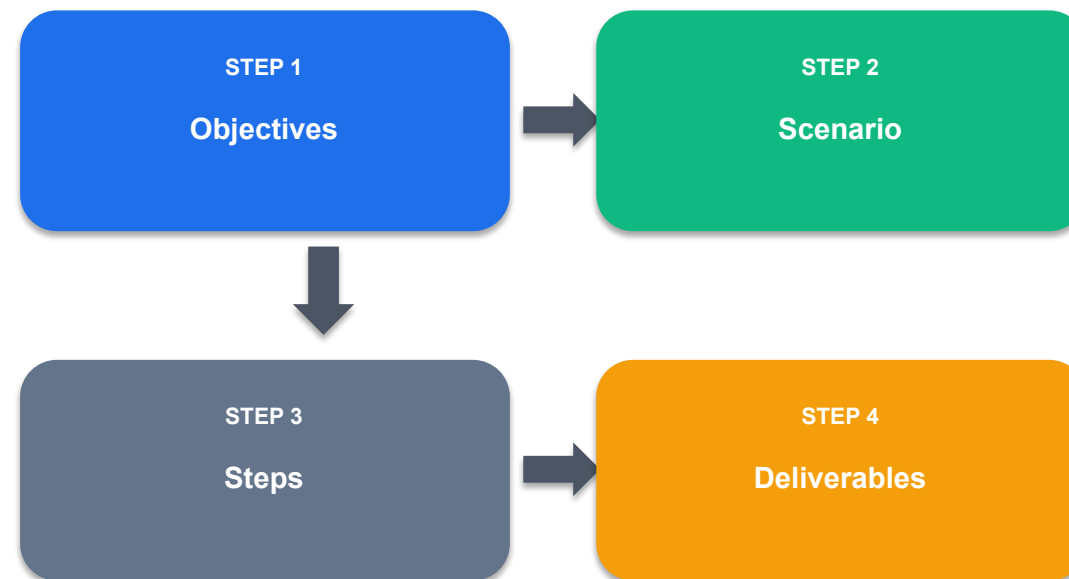
# Lab 7 - MLOps Pipelines, CI/CD, and Workflow Orchestration

Domain 3 - Deployment and Orchestration of ML Workflows

Day 2

labs/lab-07-mlops-pipelines-cicd-orchestration.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

## Lab 7 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint

## Lab 7 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

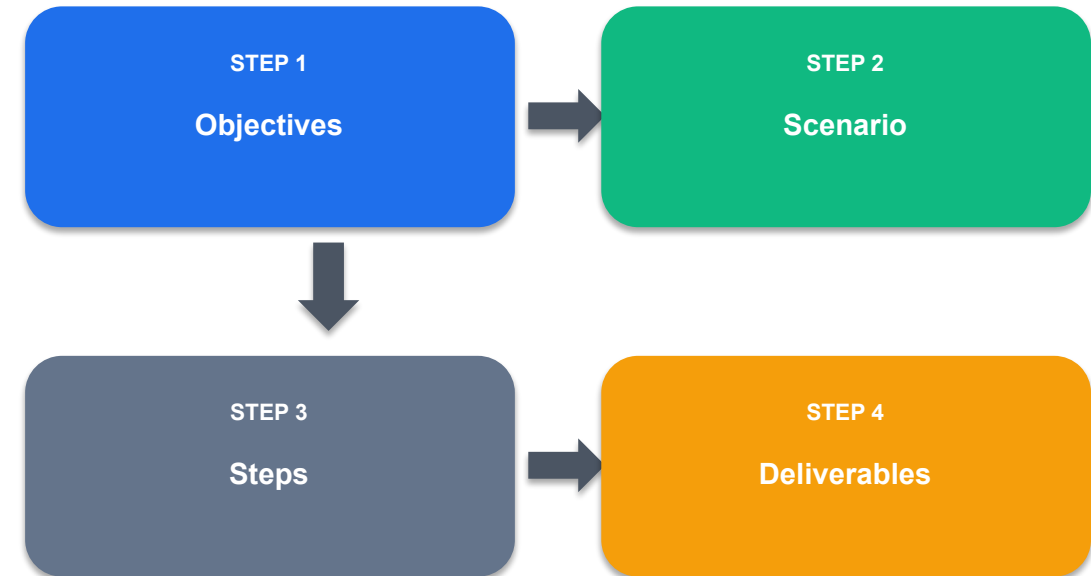
# Lab 8 - Monitoring, Security, Cost Optimization, and Exam Review

Domain 3 - Deployment and Orchestration of ML Workflows

Day 2

labs/lab-08-monitoring-security-cost-exam-review.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

## Lab 8 - Key Steps



Objectives



Scenario



Steps



Deliverables



Checkpoint



## Lab 8 - Verify and Clean Up

### Verification

1. Complete the verification or What you learned section in the lab guide.
2. Run the cleanup section exactly as written.

### Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Retain shared resources only when the next lab needs them.
3. Confirm no paid resource is left running.
4. Lab 8 performs the final course cleanup.

# Assessment

- Written Assessment (WA): 1 hour, open book.
- Practical Performance (PP): 1 hour, follows Labs 1-8.
- Complete assessment digital attendance and submit on the LMS.

# Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

## Digital Attendance (Mandatory)

 Complete the TRAQOM / SSG digital attendance checkpoint.

 Use the QR code provided through the LMS/TMS portal.

 Minimum 75% attendance is required.

# Thank You

-  You prepared data and developed, evaluated, deployed and monitored ML workloads on AWS.
-  You practised MLOps, security and cost controls mapped to the MLA-C01 exam domains.
-  Complete the Lab 8 cleanup checklist to avoid ongoing AWS charges.